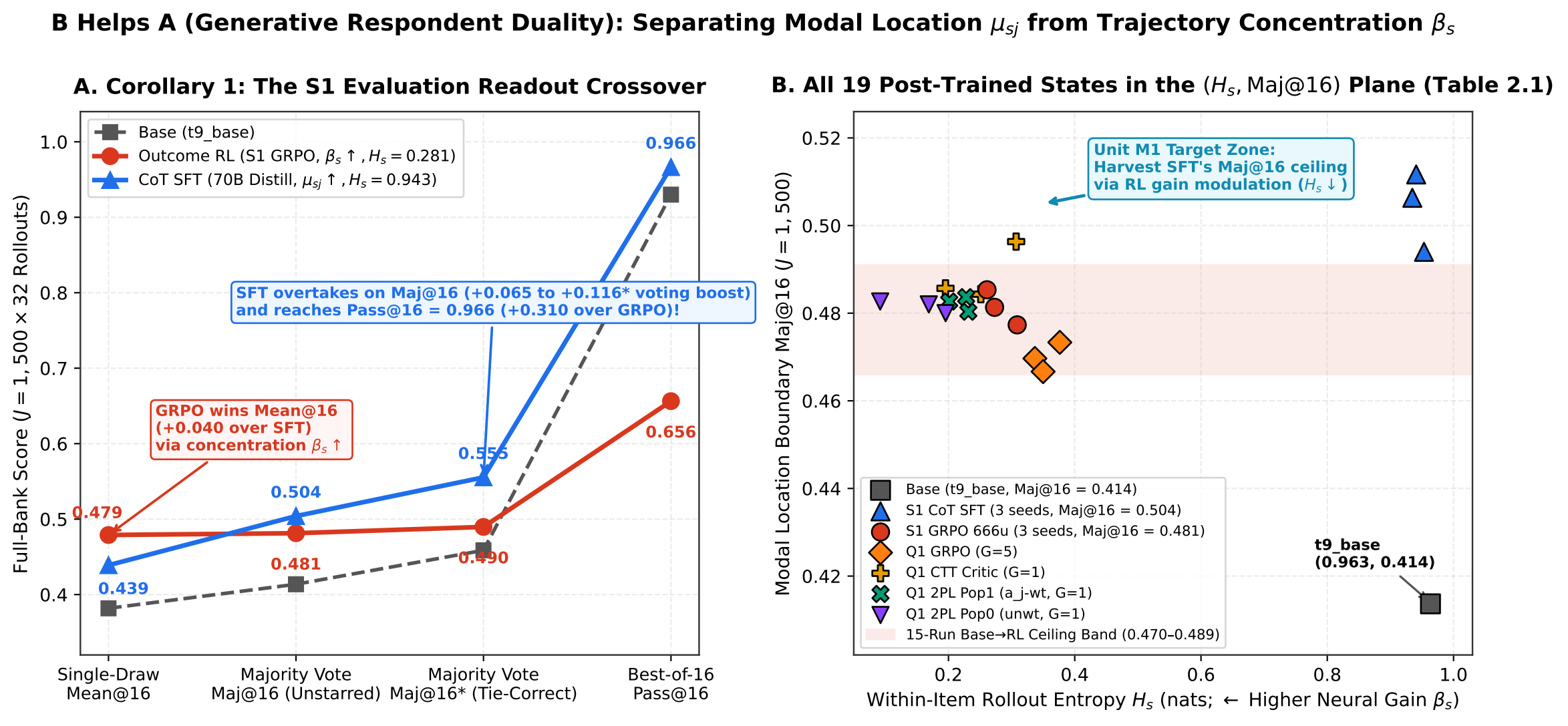
B Helps A (Generative Respondent Duality): Separating Modal Location μ_{sj} from Trajectory Concentration β_s
A. Corollary 1: The S1 Evaluation Readout Crossover
Base (t9_base)
Outcome RL (S1 GRPO, $\beta_s \uparrow$, $H_s = 0.281$)
CoT SFT (70B Distill, $\mu_{sj} \uparrow$, $H_s = 0.943$)
Full-Bank Score ($J = 1,500 \times 32$ Rollouts)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
SFT overtakes on Maj@16 (+0.065 to +0.116* voting boost)
and reaches Pass@16 = 0.966 (+0.310 over GRPO)!
GRPO wins Mean@16
(+0.040 over SFT)
via concentration $\beta_s \uparrow$
0.966
0.656
0.555
0.504
0.479
0.490
0.481
0.439
Single-Draw
Mean@16
Majority Vote
Maj@16 (Unstarred)
Majority Vote
Maj@16* (Tie-Correct)
Best-of-16
Pass@16
B. All 19 Post-Trained States in the (H_s, Maj@16) Plane (Table 2.1)
Unit M1 Target Zone:
Harvest SFT's Maj@16 ceiling
via RL gain modulation ($H_s \downarrow$)
Modal Location Boundary Maj@16 ($J = 1,500$)
0.52
0.50
0.48
0.46
0.44
0.42
Base (t9_base, Maj@16 = 0.414)
S1 CoT SFT (3 seeds, Maj@16 = 0.504)
S1 GRPO 666u (3 seeds, Maj@16 = 0.481)
Q1 GRPO (G=5)
Q1 CTT Critic (G=1)
Q1 2PL Pop1 (a_j-wt, G=1)
Q1 2PL Pop0 (unwt, G=1)
15-Run Base→RL Ceiling Band (0.470–0.489)
t9_base
(0.963, 0.414)
0.2
0.4
0.6
0.8
1.0
Within-Item Rollout Entropy H_s (nats; ← Higher Neural Gain β_s)